

Title: DCP-FRSA2025: **Mutli-Agent AI System (MAS) for Roof Design**

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Mutli-Agent AI System (MAS) for Roof Design

1. Narrative

1.1 Abstract:

Designers, engineers, and building officials are increasingly under pressure to expedite their design reviews of buildings and infrastructures. Roof assemblies and rooftop structures are critical parts of such code compliance evaluations. However, checking whether a proposed roof design complies with numerous building and local requirements is laborious and may take a student or young professionals and short-staffed city and county building departments many long time to complete [1,2,3, and 4]. For instance, the requirements for roof and rooftop structures are covered in two building codes in Florida, namely the Residential and Commercial building codes. The complexity of the topic requires increasingly deep and specific subject matter expertise, while leaps in insight may still arise from broad knowledge bridging across disciplines. Furthermore, students in Architecture, Engineering, and Construction (AEC) are faced with a breadth and depth conundrum of regulatory conformance requirements. This research aims to develop an AI framework based on a multi-agent AI system to assist students and professionals in designing and checking regulatory compliance requirements for roof and rooftop structures. The proposed MAS is devised to act as a helpful assistant and collaborator to AEC students and professional and to help building officials accelerate design reviews for roof and rooftop structures. It represents an AI Co-Roof Designer. The system's design incorporates data preprocessing, analysis, results visualization and validation. The study objectives include i) data collection, ii) data preprocessing agent development, iii) data analysis agent development, and iv) evaluation and visualization agent development. The key contributions encompasses: (1) a multi-agent architecture with an asynchronous task execution framework for flexible compute scaling for roofing design and construction; (2) a user-friendly interface to interact with the system to address educational and professional needs.

1.2 Technical Process Description

1.2.1 Research

This study centers on leveraging Machine Learning (ML) platforms, such as YOLO v11 and Mask R-CNN, Large Language Models (LLMs), and Retrieval Augmented Generation (RAG) to develop MAS for roof design Design Regulatory Conformance checking and aids in the design and construction process. The proposed system represents an AI Co-Roof Designer. The system employs specialized agents — Preprocessing, Analysis, Meta-review (which provides high level analysis) — to continuously generate, and evolve to provide better outcomes. Feedback from the output enables iterative improvement, creating a self-improving loop towards novel and high-quality outputs. This work does not aim to completely automate the roof design process with AI. Instead, the Co-Roof Designer is purpose-built for a “designer-in-the-loop” collaborative paradigm, to help AEC studetns and professional augment their design generation process and guide the design exploration in the optimum direction. Figure 1 depicts the components of the proposed AI system. These components are described in the following sections.

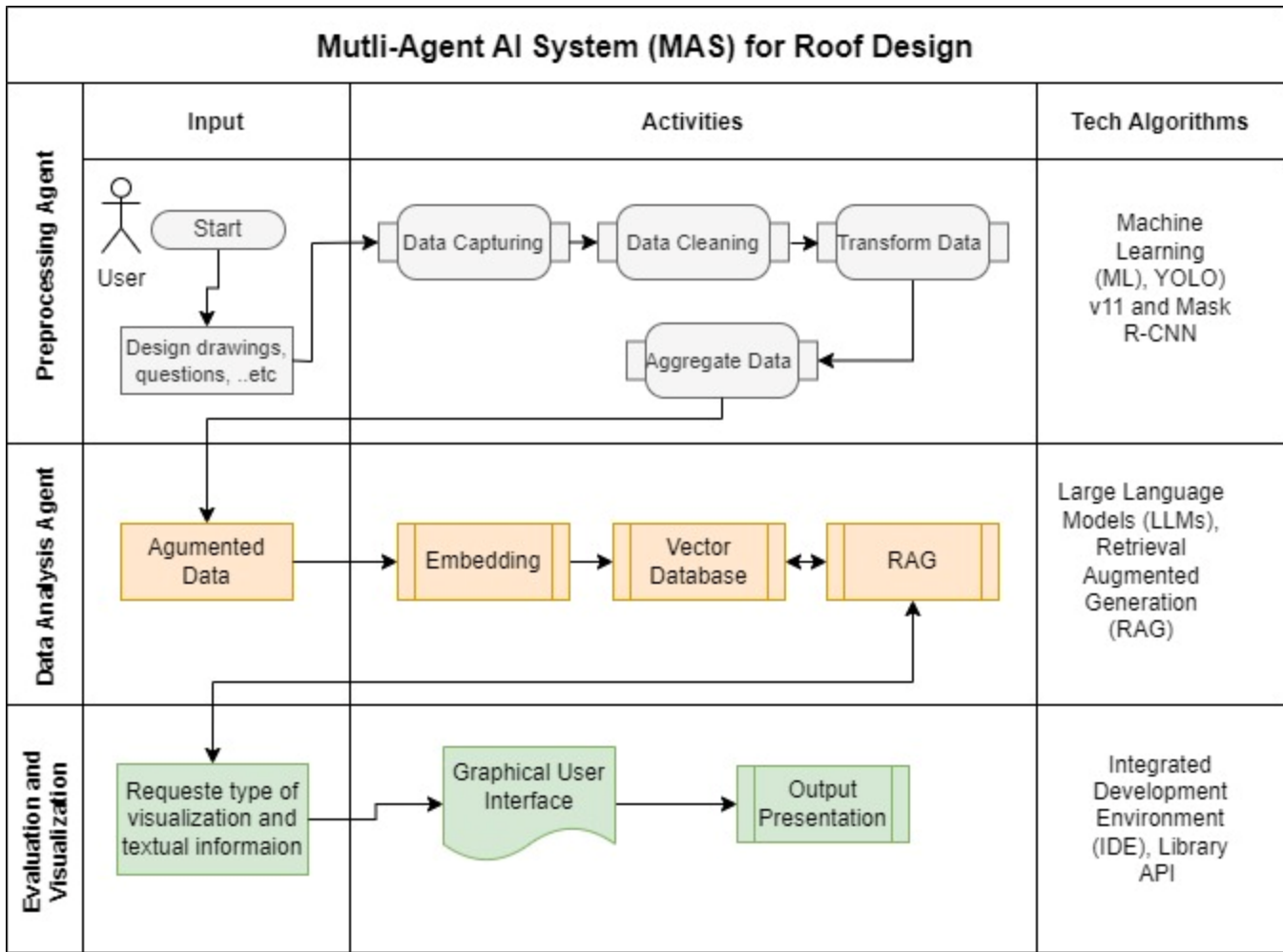
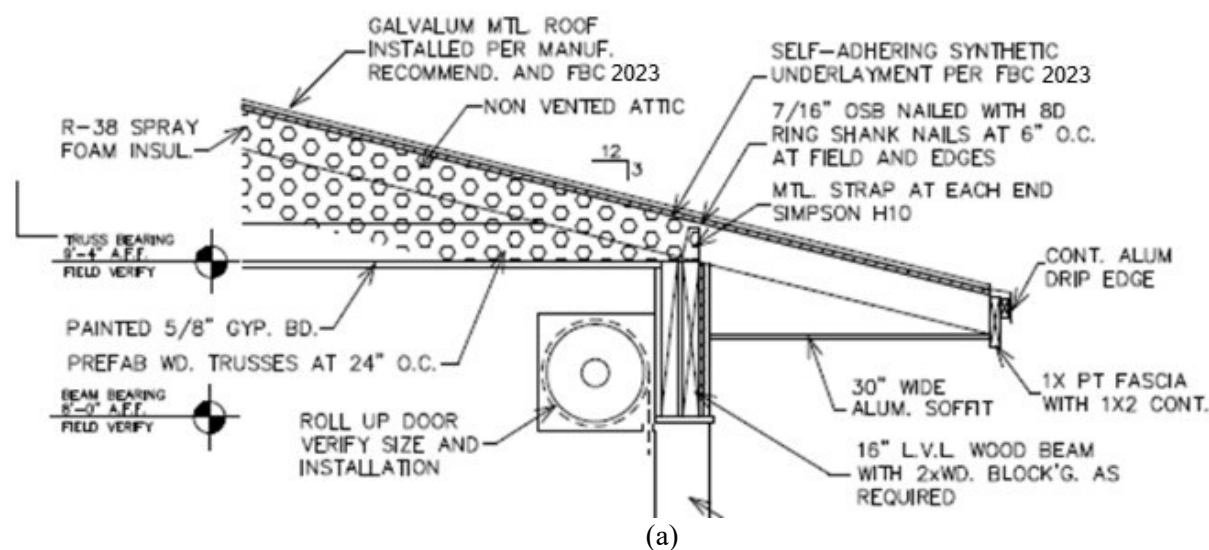


Figure 1. The different components of the AI Co-Roof designer, multi-agent system and its interaction paradigm with a designer.

- (i) the data collection phase centers on selecting roof design data from different sources including the code and local regulations (see Figure 2),



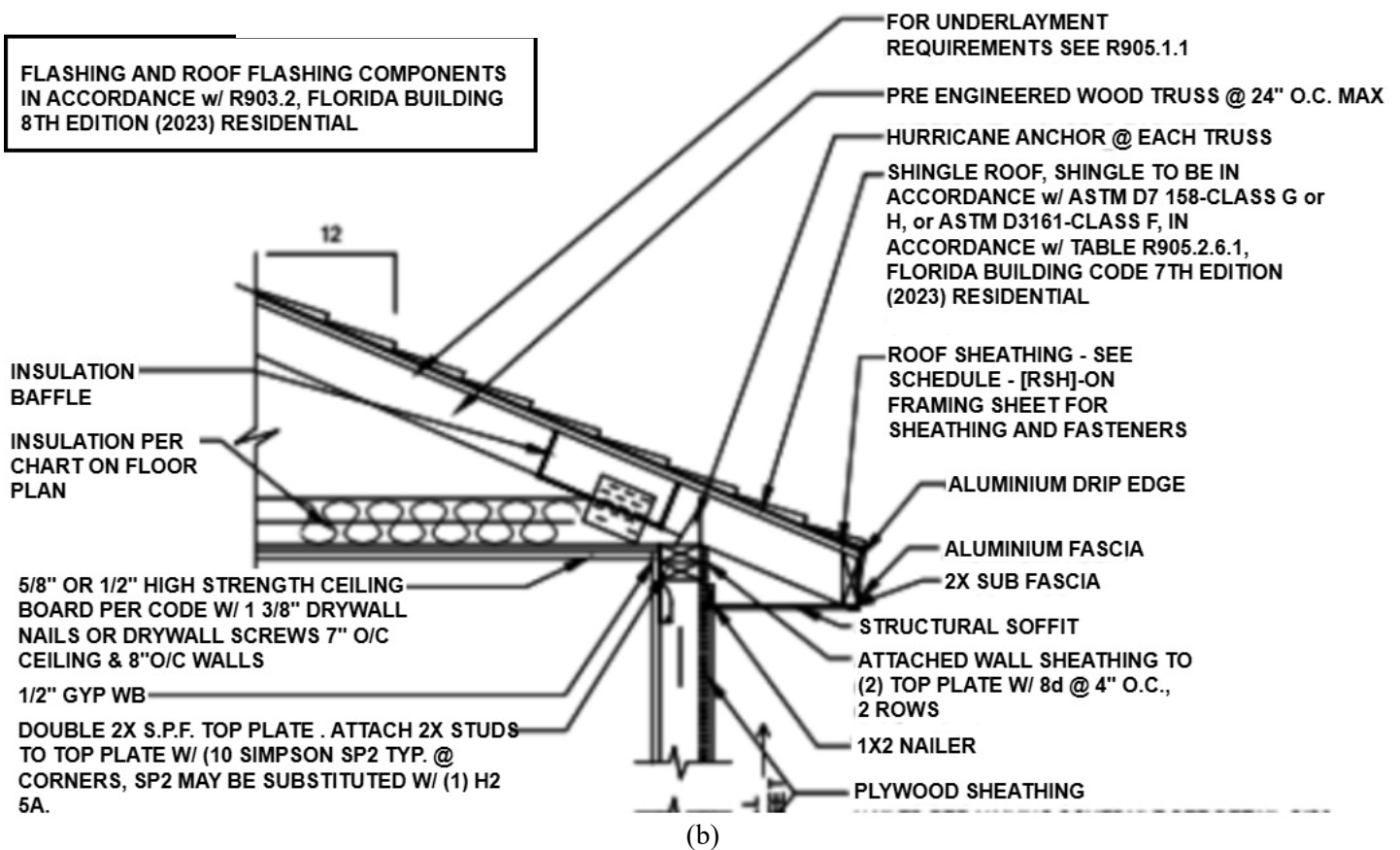


Figure 2. Examples of roof design data.

- (ii) data preprocessing agent will extract the data needed from the previous phase. This step involves leveraging Machine Learning (ML) platforms, such as a convolutional neural network called You Look Only Once (YOLO) v11 and Mask R-CNN (Mask Region-based Convolutional Neural Network) [7,8, 9, 10], to extract relevant roof design data from the PDF or image drawings (see Figure 2). This would include calibration and splitting data into training, testing, validation, and prediction (see Figure 3).

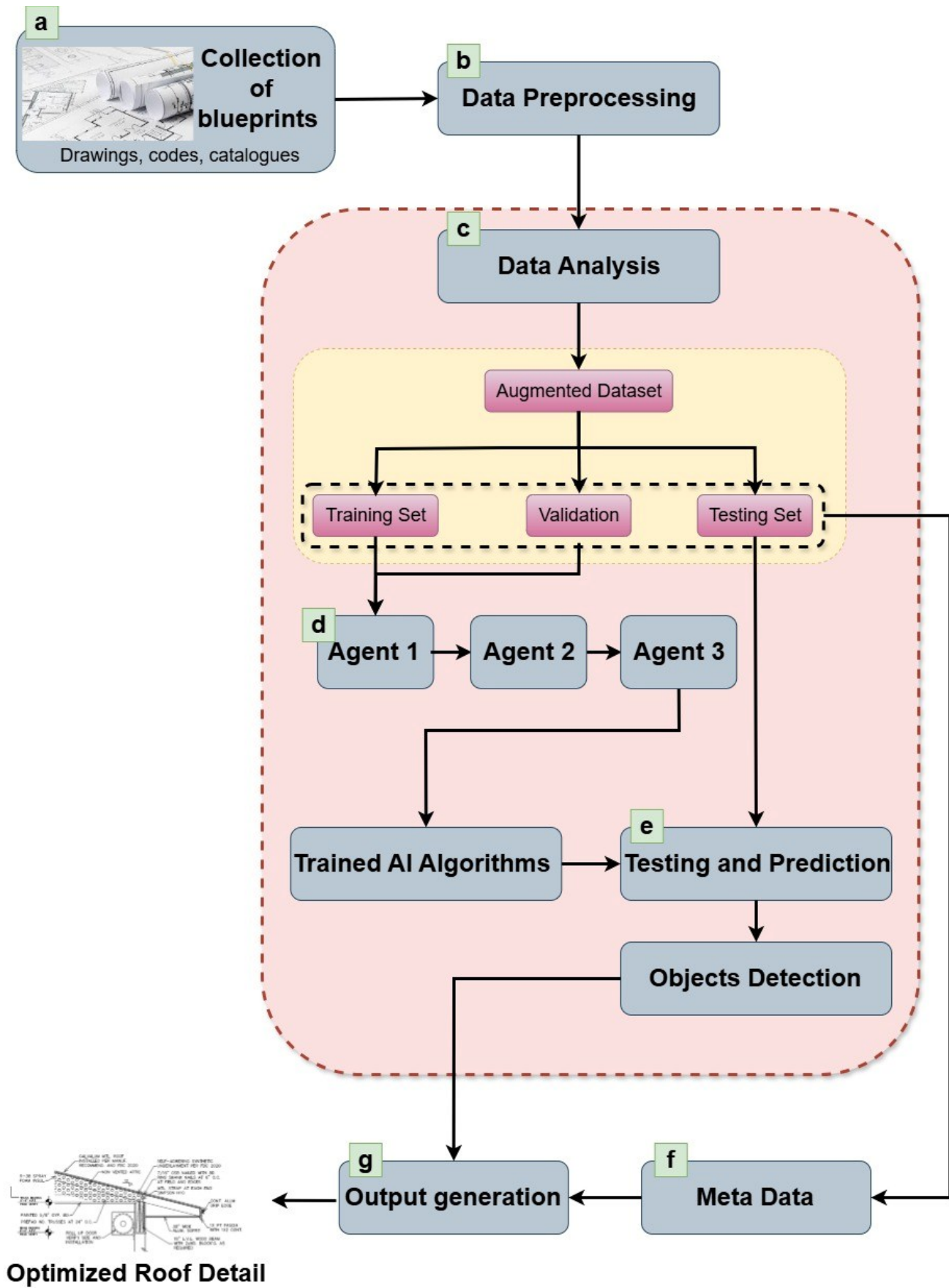


Figure 3. Outline of the subcomponents of the preprocessing agent.

- (iii) data analysis agent will analysis the data obtained and interact with the code and regulations data sources using LLMs and RAGs to provide feedback about the regulatory conformance of each roof element detected by the preprocessing agent.
- (iv) Output agent will present the result of the analysis in the requested visualization format. This could include section view of the roof with the identifying parameters and other textual information. Figure 4 depicts some examples of the proposed tool outcomes of this agent for examining roof structure for residential building in Florida.

FBC mandates that all roofing materials must meet specific ASTM standards and have Florida Product Approval to ensure compliance with wind uplift requirements.

- Asphalt Shingles:
 - Must comply with ASTM D3462 for fiberglass shingles.
 - Installation must follow manufacturer instructions and meet wind resistance ratings (e.g., ASTM D3161 Class F or ASTM D7158 Class H for high-wind zones).
 - Only allowed on roofs with a slope of 2:12 or greater.
 - Must be fastened using corrosion-resistant nails; staples are not permitted.
- Clay & Concrete Tiles:
 - Installation must adhere to FRSA/TRI Florida High Wind Tile Installation Manual.
 - Tiles must pass wind uplift tests such as TAS 101/102 in high-velocity hurricane zones (HVHZ).
 - Minimum roof slope: 2.5:12 (about 21%).
 - Fastened with screws, two nails, or approved adhesives to resist hurricane winds.
- Metal Roofing:
 - Includes standing seam, lap-seam panels, and metal shingles.
 - Must be corrosion-resistant (e.g., aluminum, galvanized steel) and tested for high-wind performance using UL 580 or ASTM E1592. Minimum slope requirements:
 - 3:12 for metal panels without sealed seams.
 - 0.5:12 for panels with sealed seams.
 - 0.25:12 for standing

Fastening & Underlayment Requirements

Underlayment must be nailed with cap fasteners to prevent wind damage. Re-nailing of roof sheathing is required during re-roofing to ensure the roof deck's integrity. Flashing and drip edge installation must follow FBC guidelines to prevent water intrusion.

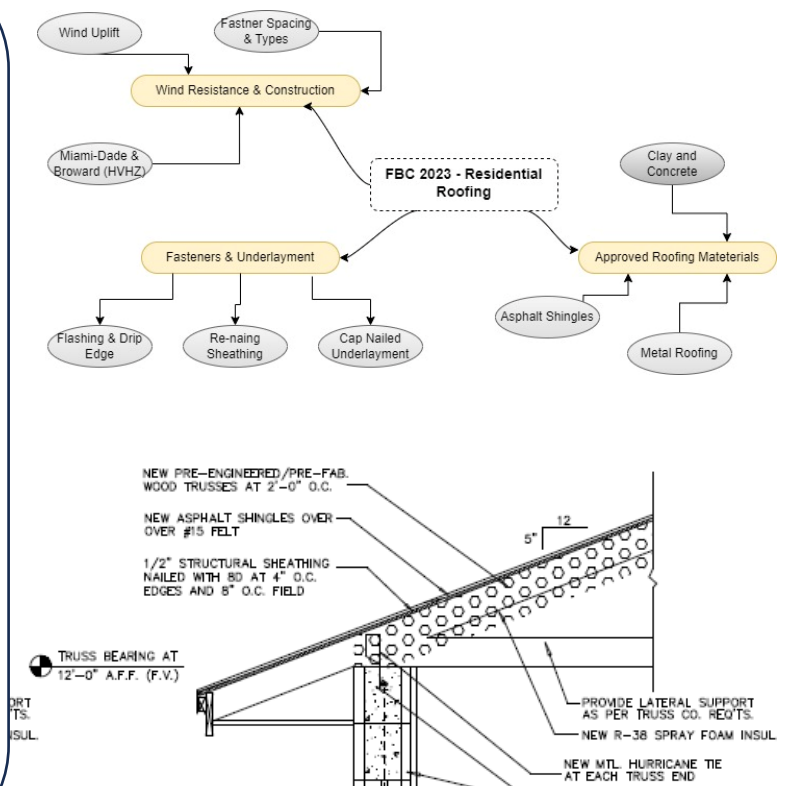


Figure 4. Output examples.

1.2.2 Teaching

The findings of this research will significantly enhance the teaching of various courses within the college by deepening students' understanding of roofing design and construction. Students in the DCP program will gain valuable insights into how Artificial Intelligence (AI) techniques can support and optimize roof design and construction processes.

Additionally, the research outcomes will contribute to the improvement of teaching materials for the Integrated Building Technology (IBT1, IBT2, IBT3, and IBT4) series and design studios, which architecture students engage with